

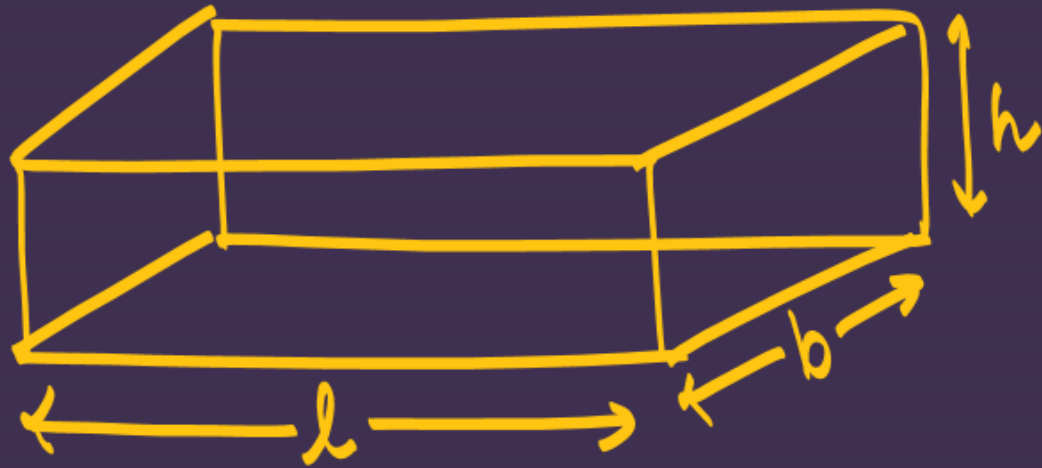
#14Days14Chapters

DAY 6

[Aim: 100/100 in Maths]

SURFACE AREAS AND VOLUMES

#Cuboid:



$$\underline{\underline{LSA}} :- 2lh + 2bh$$

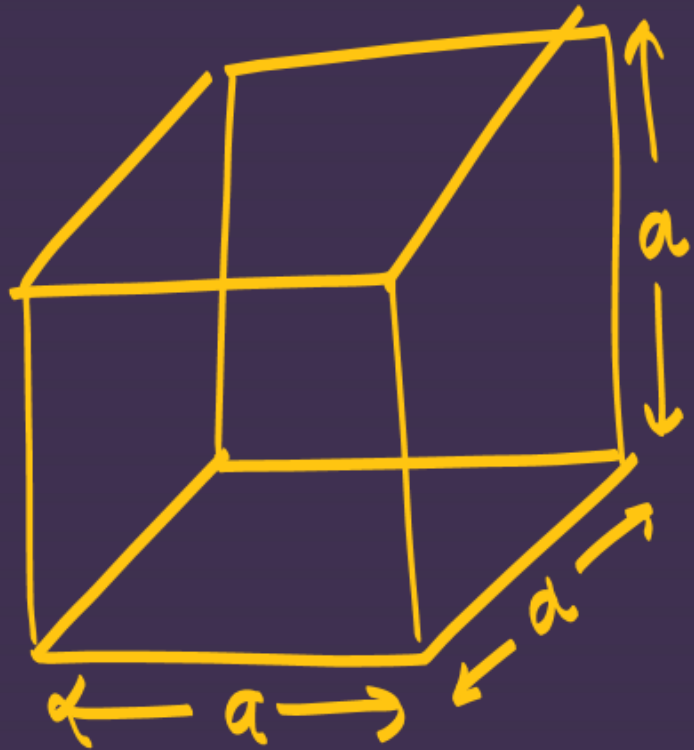
$$\underline{\underline{LSA}} \Rightarrow 2(l+b)h$$

$$\underline{\underline{TSA}} \Rightarrow 2lb + 2bh + 2hl$$

$$\underline{\underline{TSA}} \Rightarrow 2(lb + bh + hl)$$

$$\underline{\underline{Volume}} \Rightarrow l \times b \times h$$

#Cube:

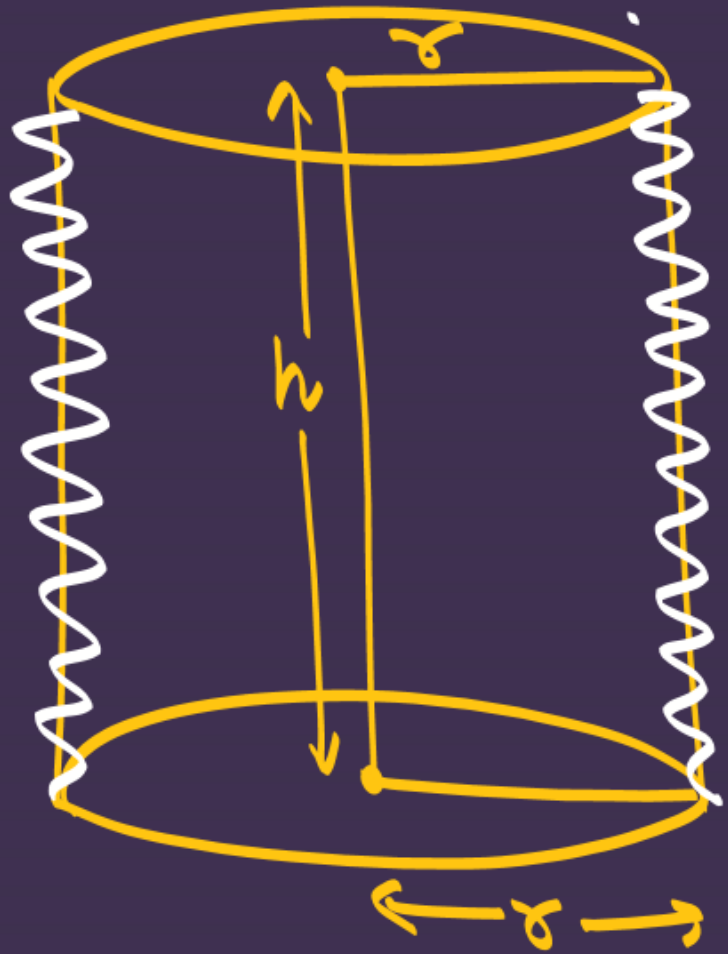


LSA $\Rightarrow 4a^2$

TSA $\Rightarrow 6a^2$

Volume $\Rightarrow a^3$

#Cylinder:

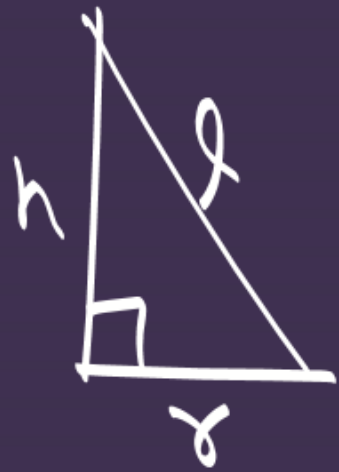


→ CSA $\Rightarrow 2\pi r h$

→ TSA $\Rightarrow 2\pi r h + 2\pi r^2$
 TSA $= 2\pi r (r + h)$

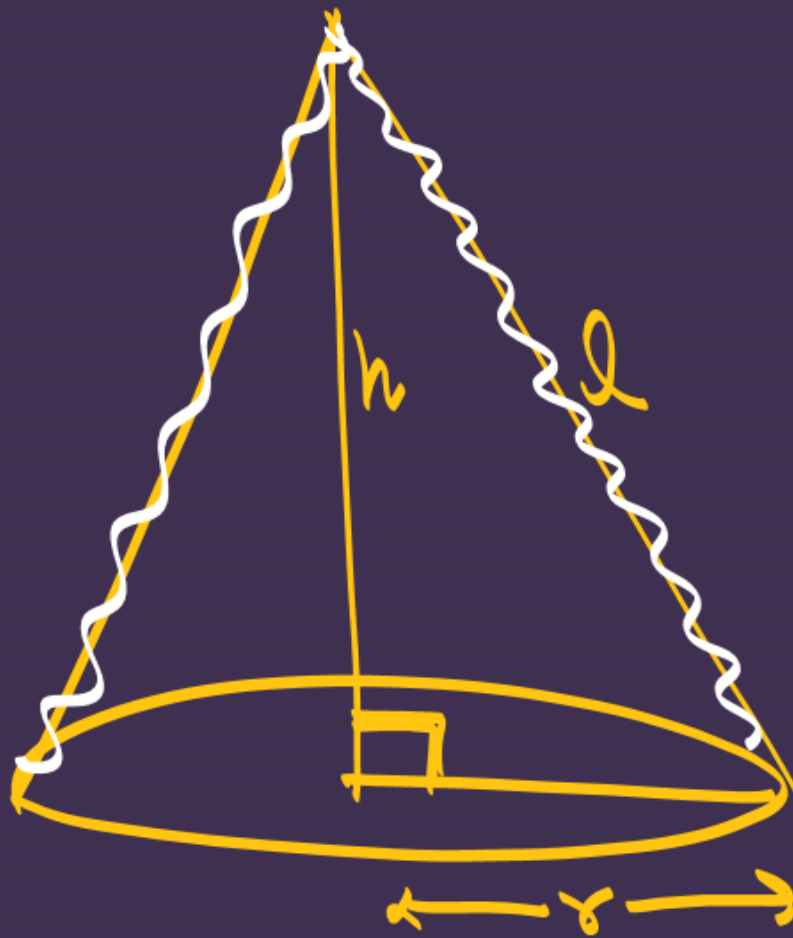
→ Volume $\Rightarrow \pi r^2 h$

#Right Circular Cone:



$$l^2 = h^2 + r^2$$

$$l = \sqrt{h^2 + r^2}$$



$$\underline{\text{CSA}} \Rightarrow \pi r l$$

$$\underline{\text{TSA}} \Rightarrow \pi r l + \pi r^2 \Rightarrow \pi r (l + r)$$

$$\underline{\text{Volume}} \Rightarrow \frac{1}{3} \pi r^2 h$$

Sphere



CSA

$$4\pi r^2$$

TSA

$$4\pi r^2$$

Vol.

$$\frac{4}{3}\pi r^3$$



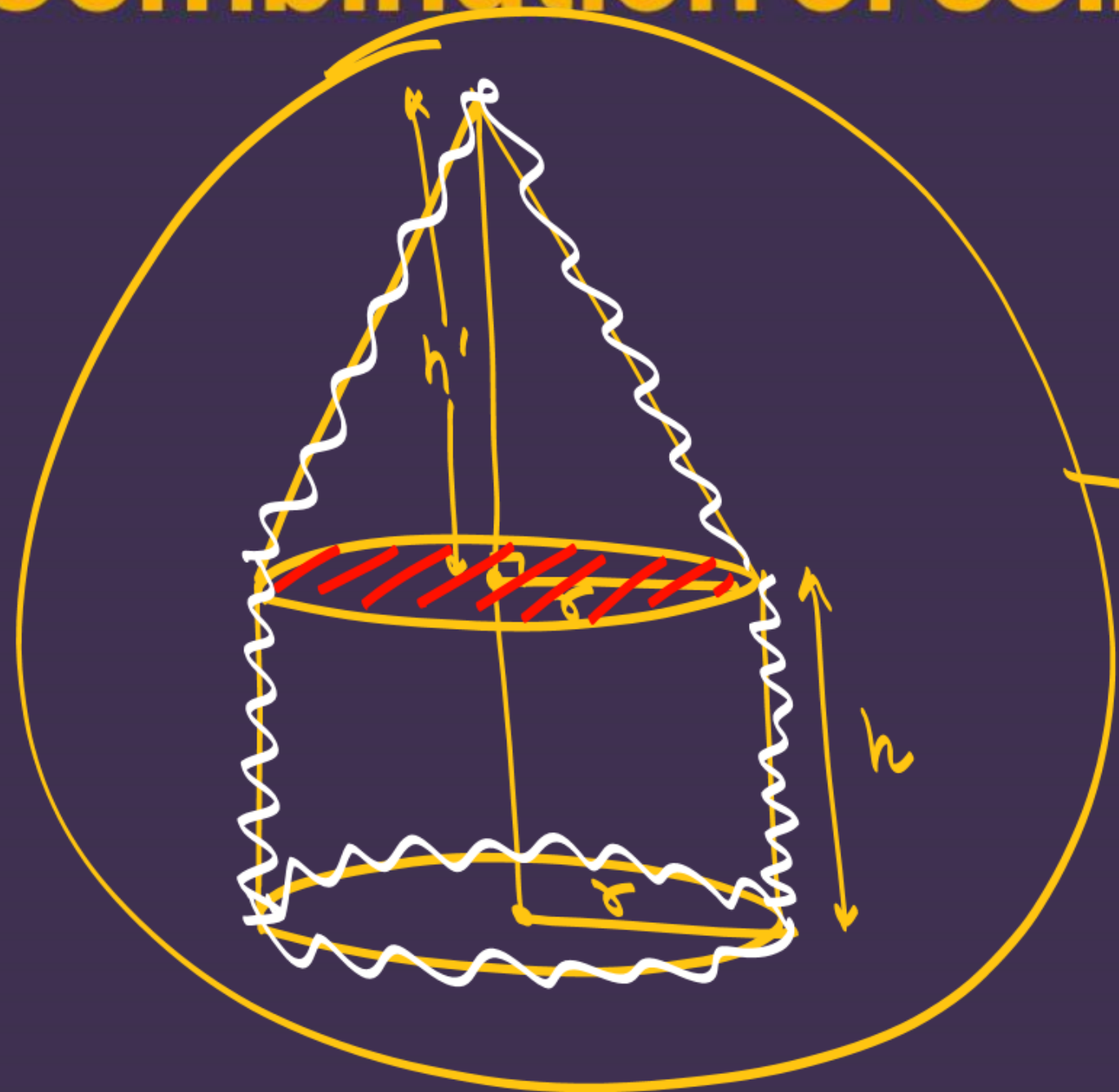
Hemi-sphere

$$2\pi r^2$$

$$2\pi r^2 + \pi r^2 \Rightarrow 3\pi r^2$$

$$\frac{1}{2} \times \frac{4}{3}\pi r^3 \Rightarrow \boxed{\frac{2}{3}\pi r^3}$$

#Combination of Solids:

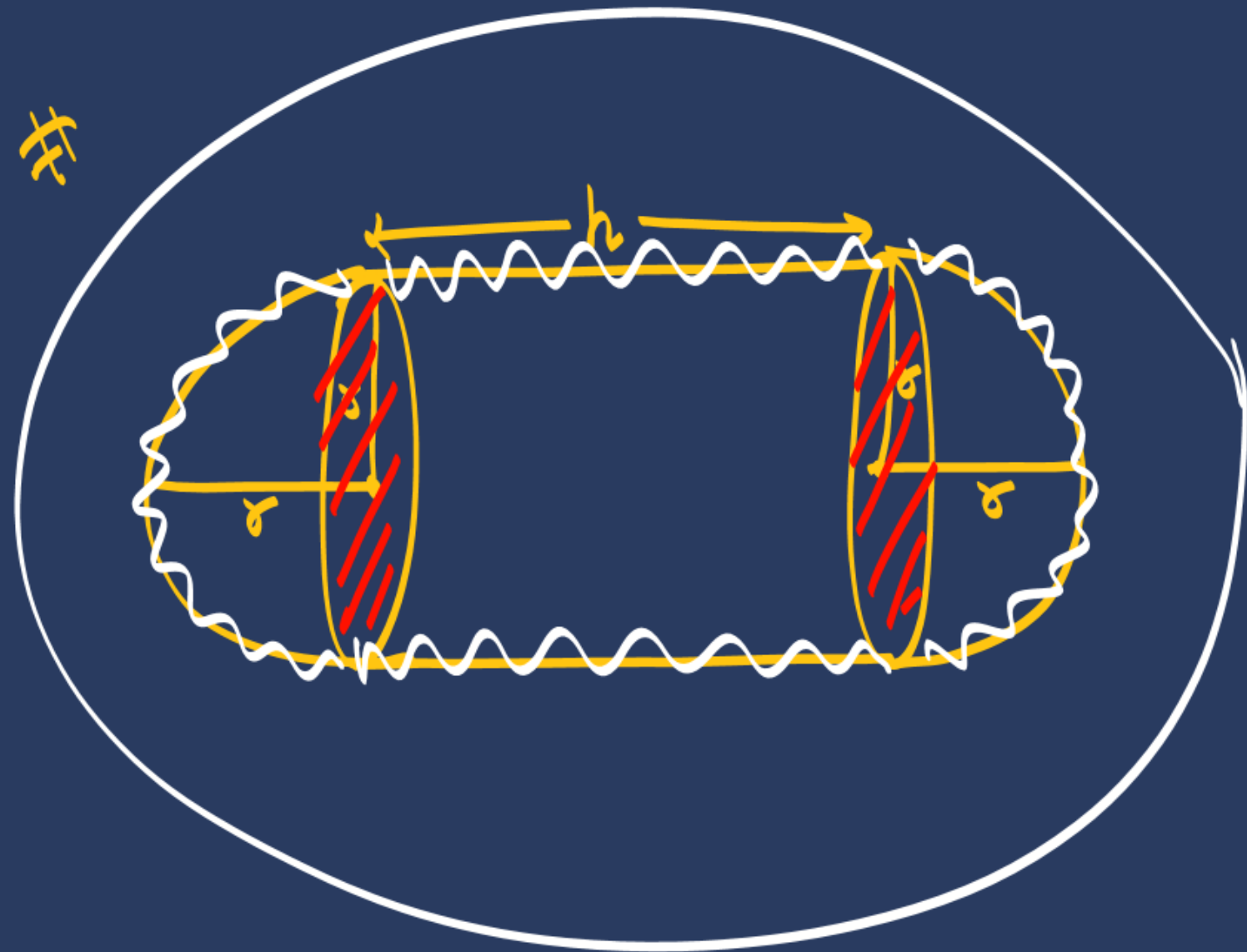


SA $\Rightarrow$ CSA cone + CSA cylinder
+ base circle
 $\Rightarrow [\pi r l + 2\pi r h + \pi r^2]$

$$l = \sqrt{h'^2 + r^2}$$

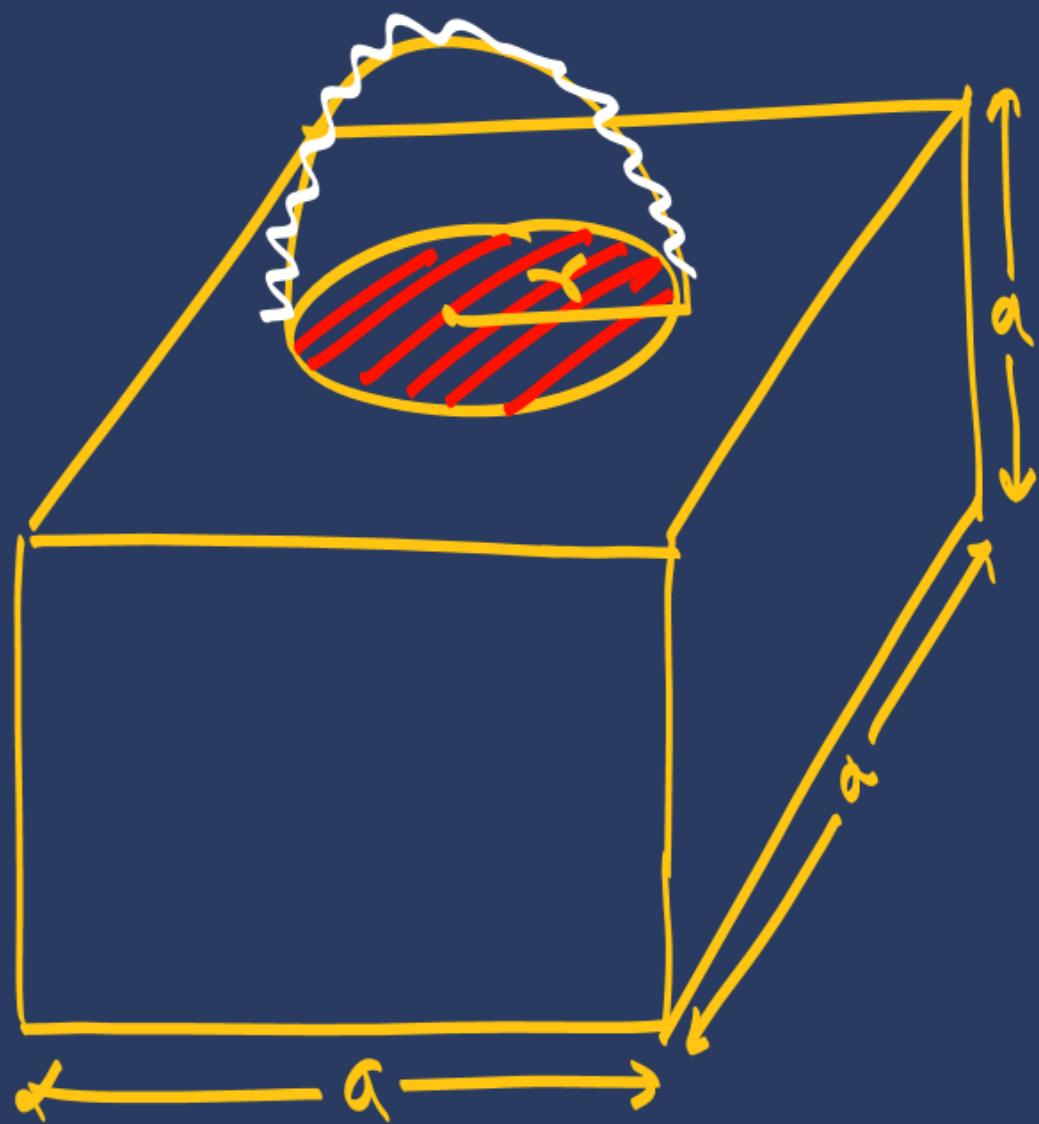
Vol $\Rightarrow$ Vol. cone + Vol. cylinder
 $\Rightarrow \underline{\left[\frac{1}{3} \pi r^2 h' + \pi r^2 h \right]}$

#



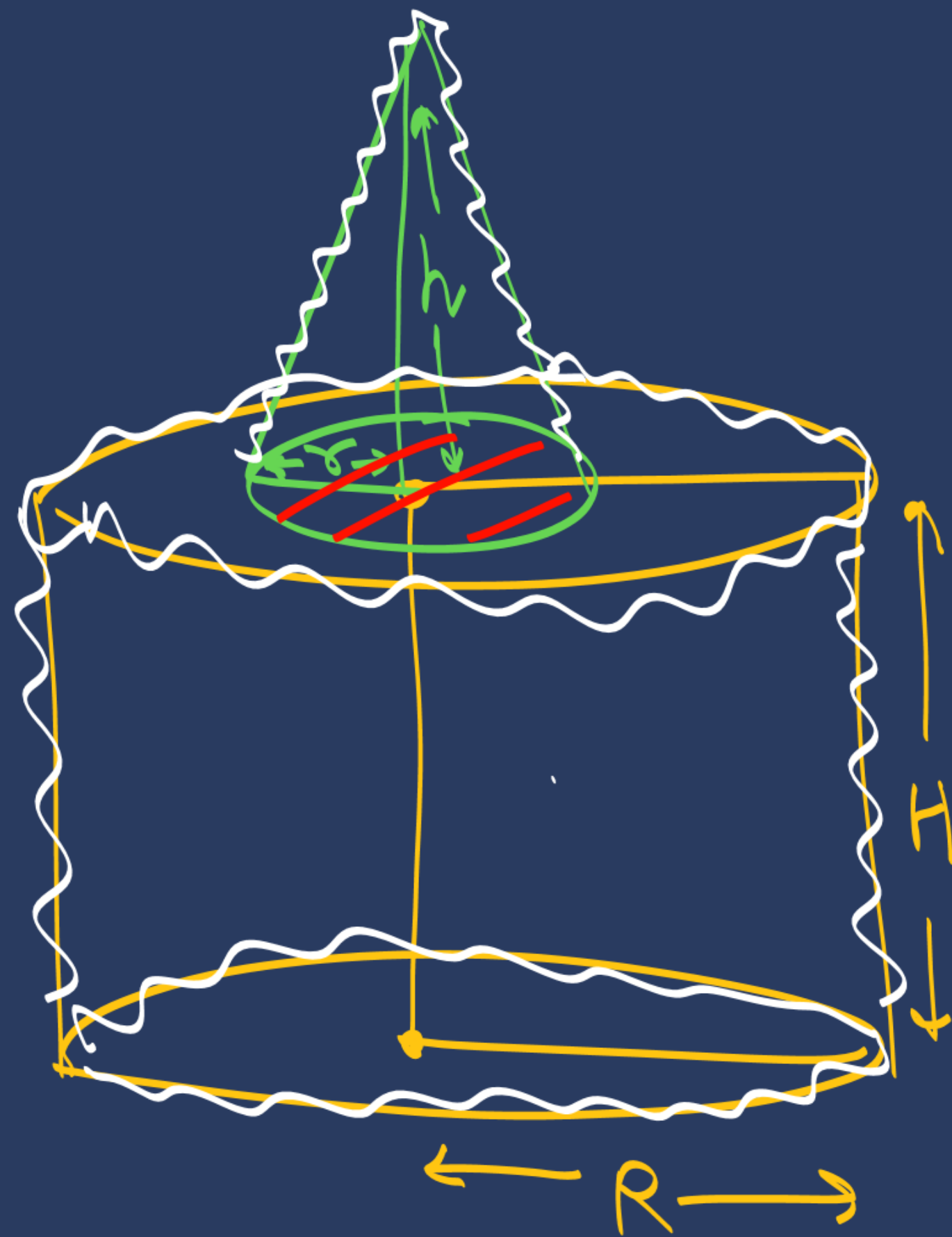
→ $SA \Rightarrow CSA_{\text{cylinder}} + 2 \times CSA_{\text{Hem.}}$
 $\Rightarrow \underline{\underline{[2\pi r h + 2 \times (2\pi r^2)]}}$

→ $Vol. \Rightarrow Vol._{\text{cylinder}} + 2 \times Vol._{\text{hemisphere}}$
 $\Rightarrow \underline{\underline{[\pi r^2 h + 2 \times (\frac{2}{3}\pi r^3)]}}$



→ SA $\Rightarrow$ CSA hem + TSA cube $- \pi r^2$
 $\Rightarrow \underline{\underline{[2\pi r^2 + 6a^2 - \pi r^2]}}$

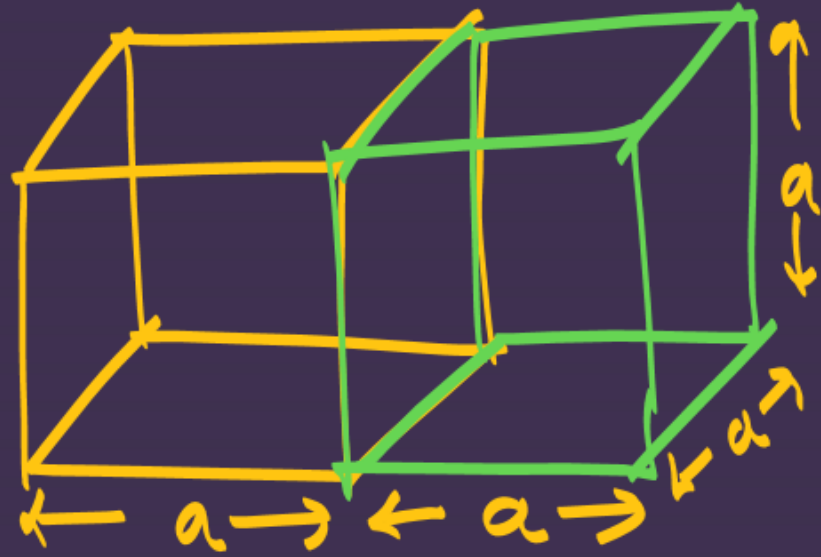
→ Vol $\Rightarrow$ Vol. hem + Vol. cube



$\underline{\underline{SA}} \Rightarrow (SA_{\text{cone}} + TSA_{\text{cylinder}} - \pi r^2)$
 $\Rightarrow \underline{\underline{\left[(\pi r l) + 2\pi R(R+H) - \pi r^2 \right]}}$

$\underline{\underline{Vol}} \Rightarrow Vol_{\text{cone}} + Vol_{\text{cylinder}}$
 $\Rightarrow \underline{\underline{\frac{1}{3}\pi r^2 h + \pi R^2 H}}$

#LP: 2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.

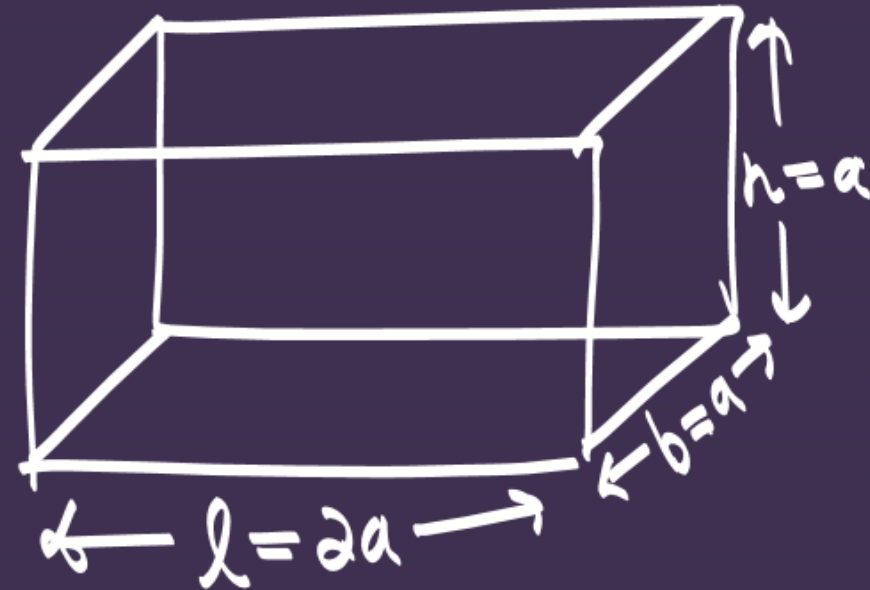


$$\text{Vol. of cube} = 64 \text{ cm}^3$$

$$a^3 = 64$$

$$a = \sqrt[3]{64}$$

$$\boxed{a = 4 \text{ cm}}$$



$$\begin{aligned} SA &\Rightarrow 2(lb + bh + hl) \\ &\Rightarrow 2(2a \cdot a + a \cdot a + a \cdot 2a) \end{aligned}$$

$$\Rightarrow 2(2a^2 + a^2 + 2a^2)$$

$$\boxed{SA = 10a^2} \Rightarrow 10 \times 4^2 \Rightarrow \underline{\underline{160 \text{ cm}^2}}$$

#LP: Volume and surface area of a solid hemisphere are numerically equal what is the diameter of the hemisphere?

[CBSE 2017]



$$(Vol)_{hem} = (TSA)_{hem}$$

$$\frac{2}{3} \pi r^3 = 3 \pi r^2$$

$$r = \frac{9}{2}$$

$$d = 2r$$
$$= 2 \times \frac{9}{2}$$

$$d = 9 \text{ cm}$$

#LP: A right circular cylinder and a cone have equal base radius and equal height. If their curved surface are in the ratio 8:5, then the radius of the base to the height are in the ratio :

- (A) 2:3
- (B) 4:3
- ☒ (C) 3:4
- (D) 3:2

$$\frac{r}{h} = ?$$

(h, r)

$$\Rightarrow \frac{(CSA)_{\text{cylinder}}}{(CSA)_{\text{cone}}} = \frac{8}{5}$$

$$\Rightarrow \frac{\cancel{2\pi r} h}{\cancel{\pi r} l} = \frac{8}{5}$$

$$\boxed{\frac{h}{l} = \frac{4}{5}}$$

$$\frac{h^2}{l^2} = \frac{4^2}{5^2}$$

$$\frac{h^2}{r^2 + h^2} = \frac{16}{25}$$

$$25h^2 = 16r^2 + 16h^2$$

$$25h^2 - 16h^2 = 16r^2$$

$$9h^2 = 16r^2$$

$$\frac{9}{16} = \frac{r^2}{h^2}$$

$$\sqrt{\frac{9}{16}} = \frac{r}{h} \Rightarrow \boxed{\frac{r}{h} = \frac{3}{4}}$$



#LP: The radius and the height of a right circular cone are in the ratio 5:12, respectively. If its volume is 314 m³, then find its slant height and the radius. [Use $\pi = 3.14$]



$$\frac{r}{h} = \frac{5}{12}$$

$$\text{Let } \begin{cases} r = 5k \\ h = 12k \end{cases}$$

$$\underline{\underline{r = 5(1) = 5\text{m}}}$$

$$\underline{\underline{h = 12(1) = 12\text{m}}}$$

$$V = 314\text{m}^3$$

$$\frac{1}{3} \pi r^2 h = 314$$

$$\frac{1}{3} \times \frac{3.14}{100} \times (5k)^2 \times (12k) = 314$$

$$\frac{1}{12} \times 5k^2 \times 12k = 1$$

$$k^3 = 1$$

$$\underline{\underline{k = 1}}$$

$$\underline{\underline{2. r = ??}}$$

$$\text{we know, } l^2 = h^2 + r^2$$

$$l = \sqrt{(12)^2 + (5)^2}$$

$$l = \sqrt{169}$$

$$\underline{\underline{l = 13\text{m}}}$$

#LP: A solid toy is in the form of a hemisphere surmounted by a right circular cone of same radius. The height of the cone is 10 cm and the radius of the base is 7 cm. Determine the volume of the toy. Also find the area of the coloured sheet required to cover the toy. (Use $\pi=22/7$ and $\sqrt{149}=12.2$).



$$\rightarrow Vol = Vol. \text{ cone} + Vol. \text{ hem}$$

$$\Rightarrow \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

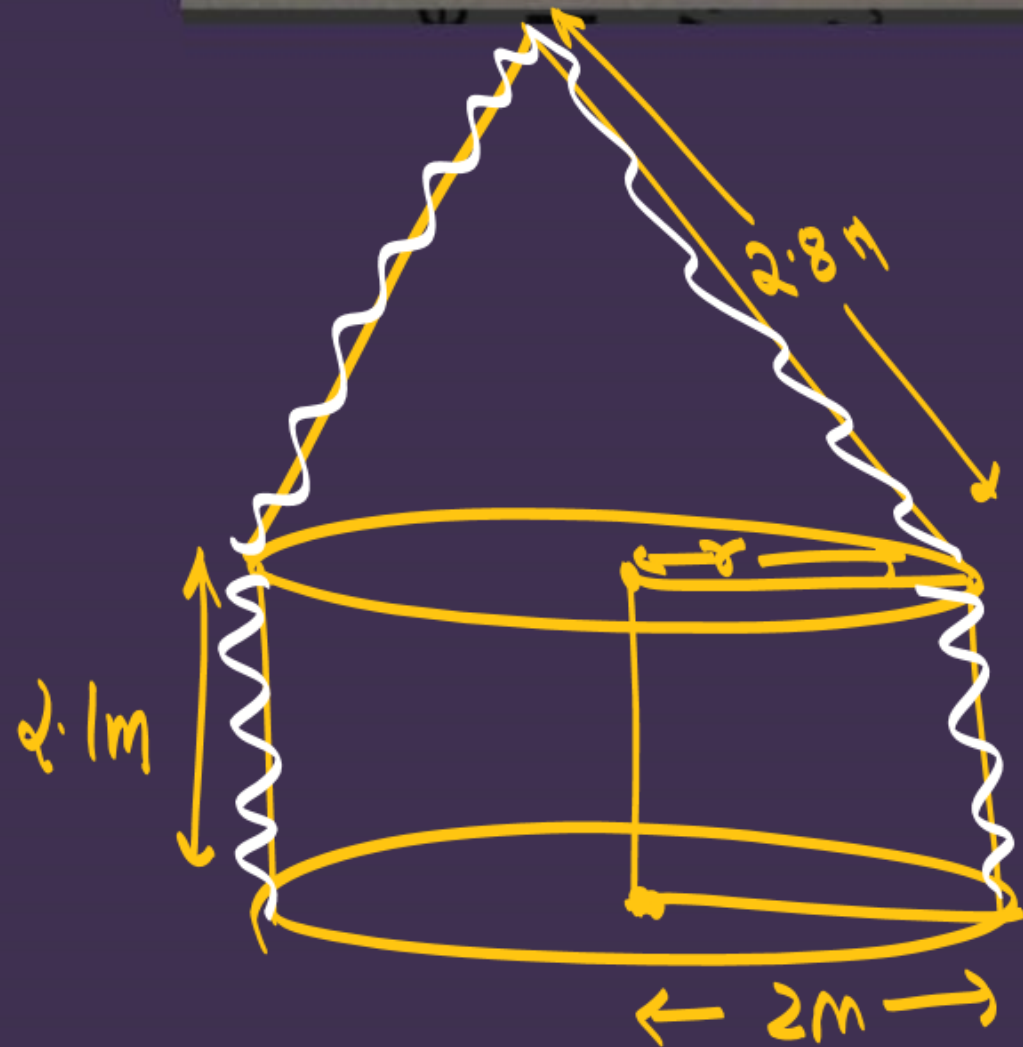
$$\Rightarrow \frac{\pi r^2}{3} (h + 2r) \Rightarrow \frac{22}{7} \times \frac{(7)^2}{3} (10 + 2(7)) \text{ cm}^3$$

(ii) SA $\Rightarrow$ CSA cone + CSA hem.

#LP:

A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1m and 4m respectively, and the slant height of the top is 2.8m, find the area of the canvas used for making the tent. Also, find the cost of canvas of the tent at the rate of Rs.500/m². (the base of the tent will not be covered with canvas)

[CBSE 2016]

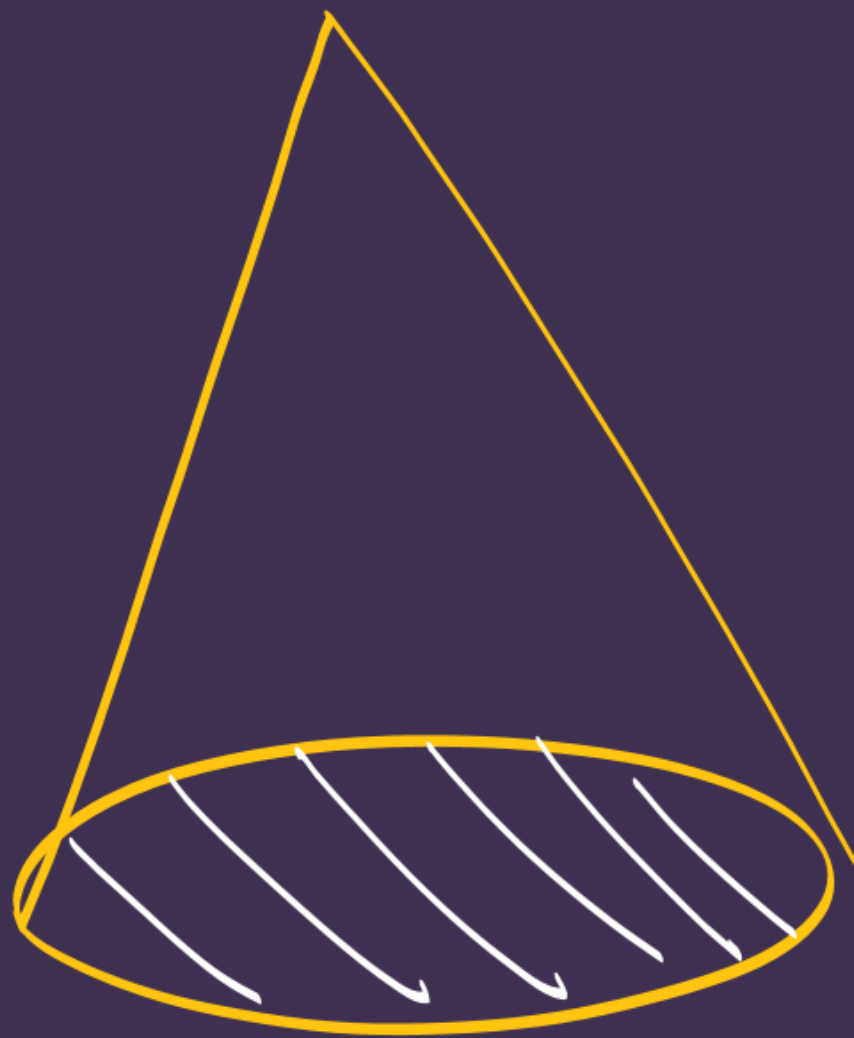


(i) Area of sheet Req $\Rightarrow$ CSA cone + CSA cylinder
 $\Rightarrow \pi r l + 2\pi r h$
 $\Rightarrow \pi r (l + 2h) \text{ cm}^2 \Rightarrow 44 \text{ m}^2$

(ii) $1 \text{ m}^2 \rightarrow ₹ 500$
 $44 \text{ m}^2 \rightarrow ₹ (44 \times 500)$

$₹ 22,000$

Q4 #LP: A conical tent is to accommodate 11 persons. Each person must have 4 sq m of the space on the ground and 20 cu m of air to breathe. Find the height of the cone.



Ground →
 1 person → 4 m^2
 11 person → 11×4
 Ar. of base → 44 m^2
 $\pi r^2 = 44\text{ m}^2$ — (I)

1 person → 20 m^3
 11 person → $11 \times 20\text{ m}^3$
 Vol. cone = 220 m^3

$$\frac{1}{3} \pi r^2 h = 220$$

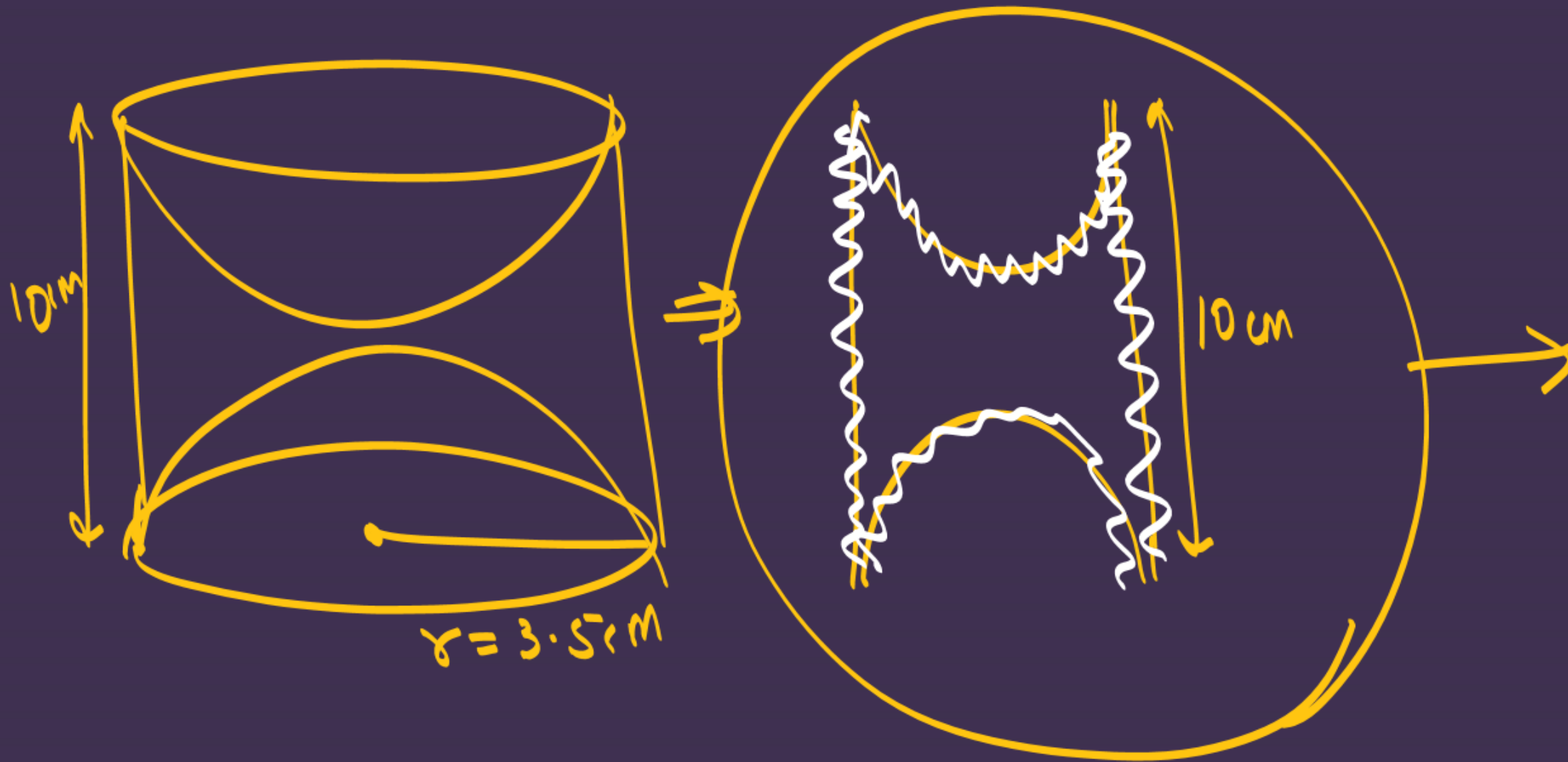
$$\rightarrow \frac{1}{3} \times 44 \times h = 220$$

$$\rightarrow h = 15\text{ m}$$

#LP:

Wooden article was made by a scooping out hemisphere from each end of a solid cylinder as shown in figure if the height of cylinder is 10 centimetre and the radius of its base is 3.5 centimetre then find the total surface area of the article .

[CBSE 2014,18]



$$\begin{aligned} TSA &= CSA_{\text{cylinder}} + 2(CSA_{\text{hemi}}) \\ &\Rightarrow 2\pi r h + 2(2\pi r^2) \\ &= \underline{\underline{2\pi r (h + 2r)}} \end{aligned}$$

#LP: A golf ball is spherical with about 300 – 500 dimples that help increase its velocity while in play. Golf balls are traditionally white but available in colours also. In the given figure, a golf ball has diameter 4.2 cm and the surface has 315 dimples (hemi-spherical) of radius 2 mm. Based on the above, answer the following questions :

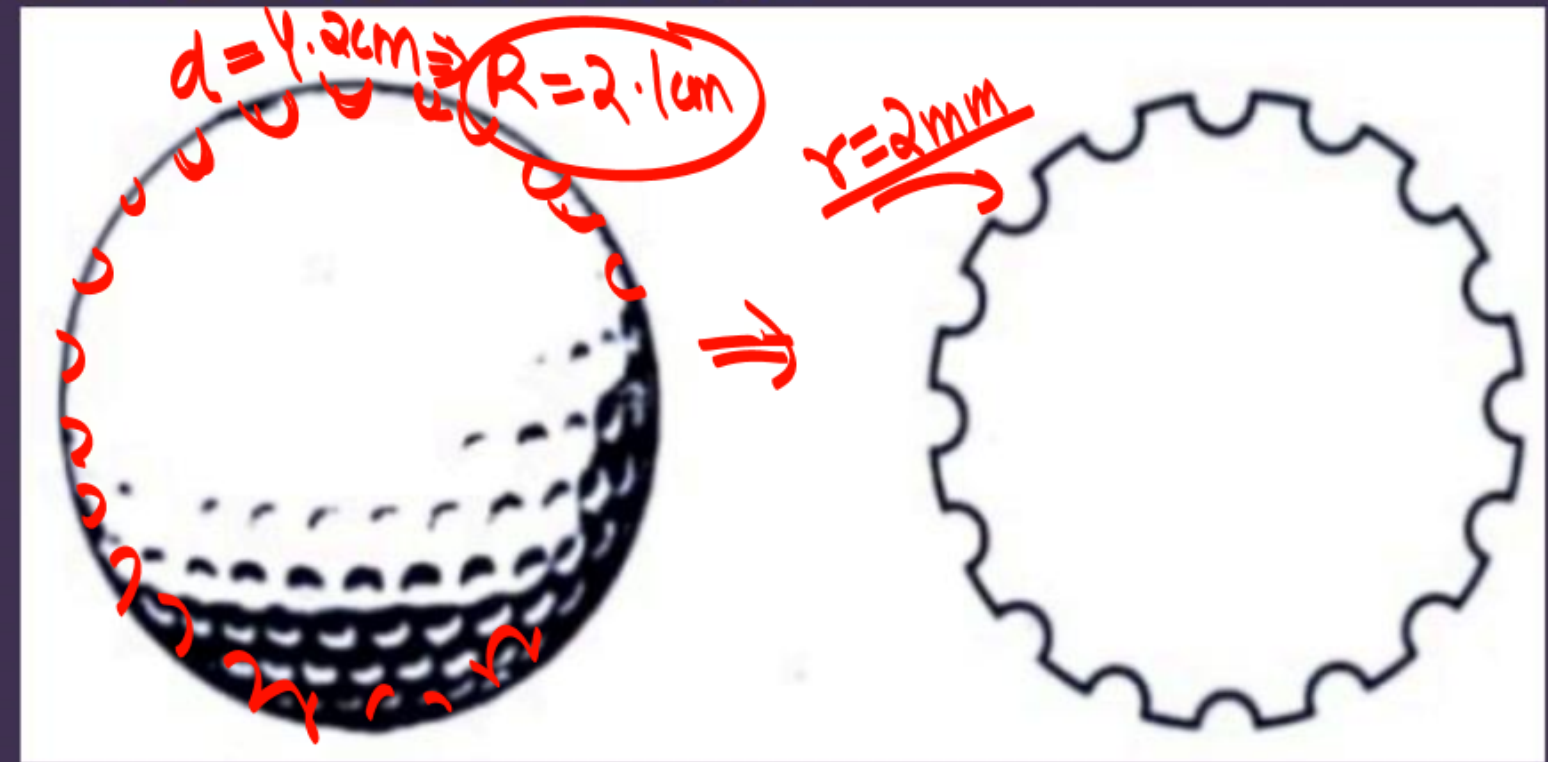
- (i) Find the surface area of one such dimple.
- (ii) Find the volume of the material dug out to make one dimple.
- (iii) (a) Find the total surface area exposed to the surroundings.
(b) Find the volume of the golf ball.

(i)

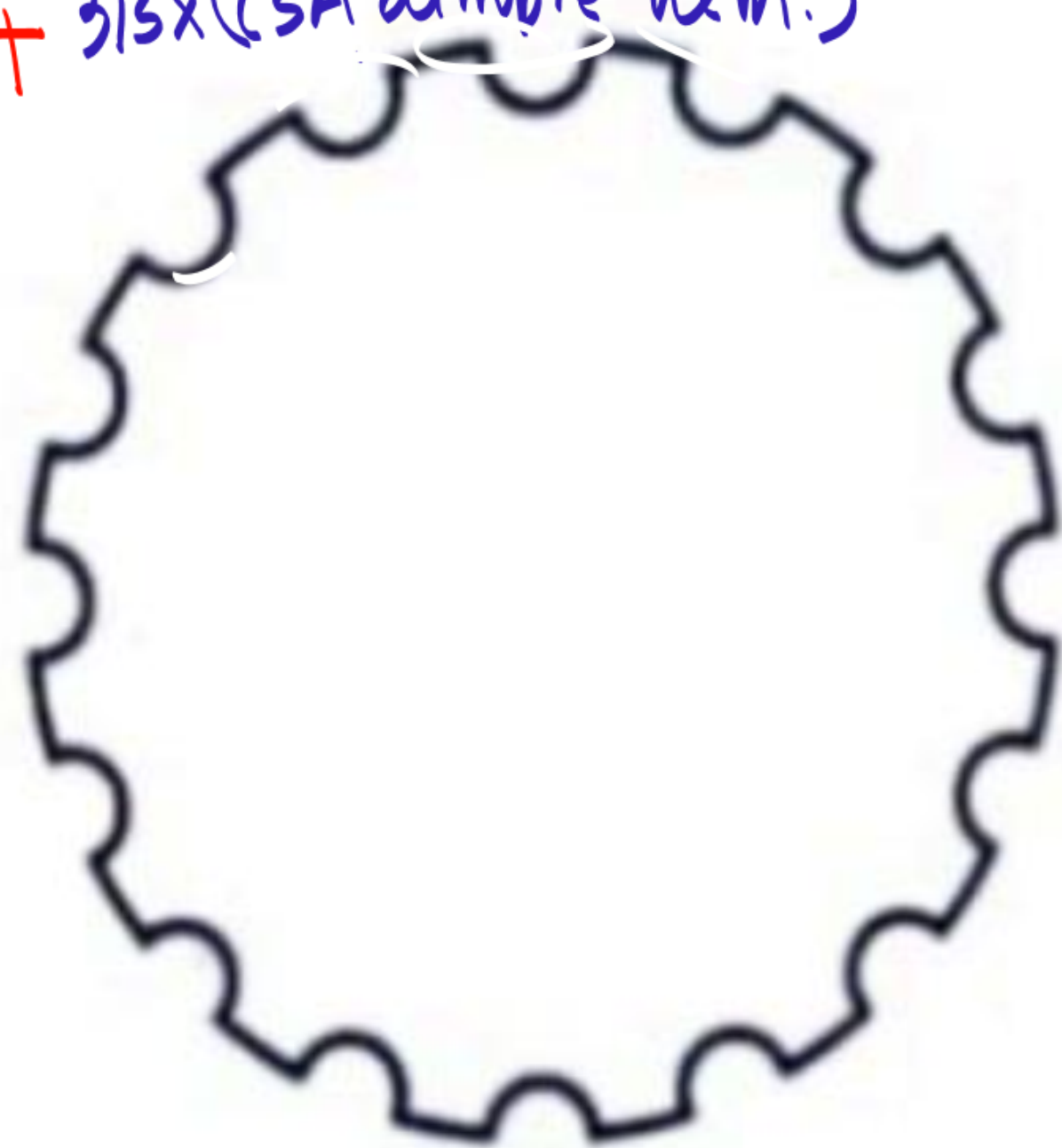
$$CSA \Rightarrow \underline{\underline{2\pi r^2 \text{ mm}^2}}$$

(ii) Vol. dug out $\Rightarrow$ Vol. of 1 dimple (hemis)

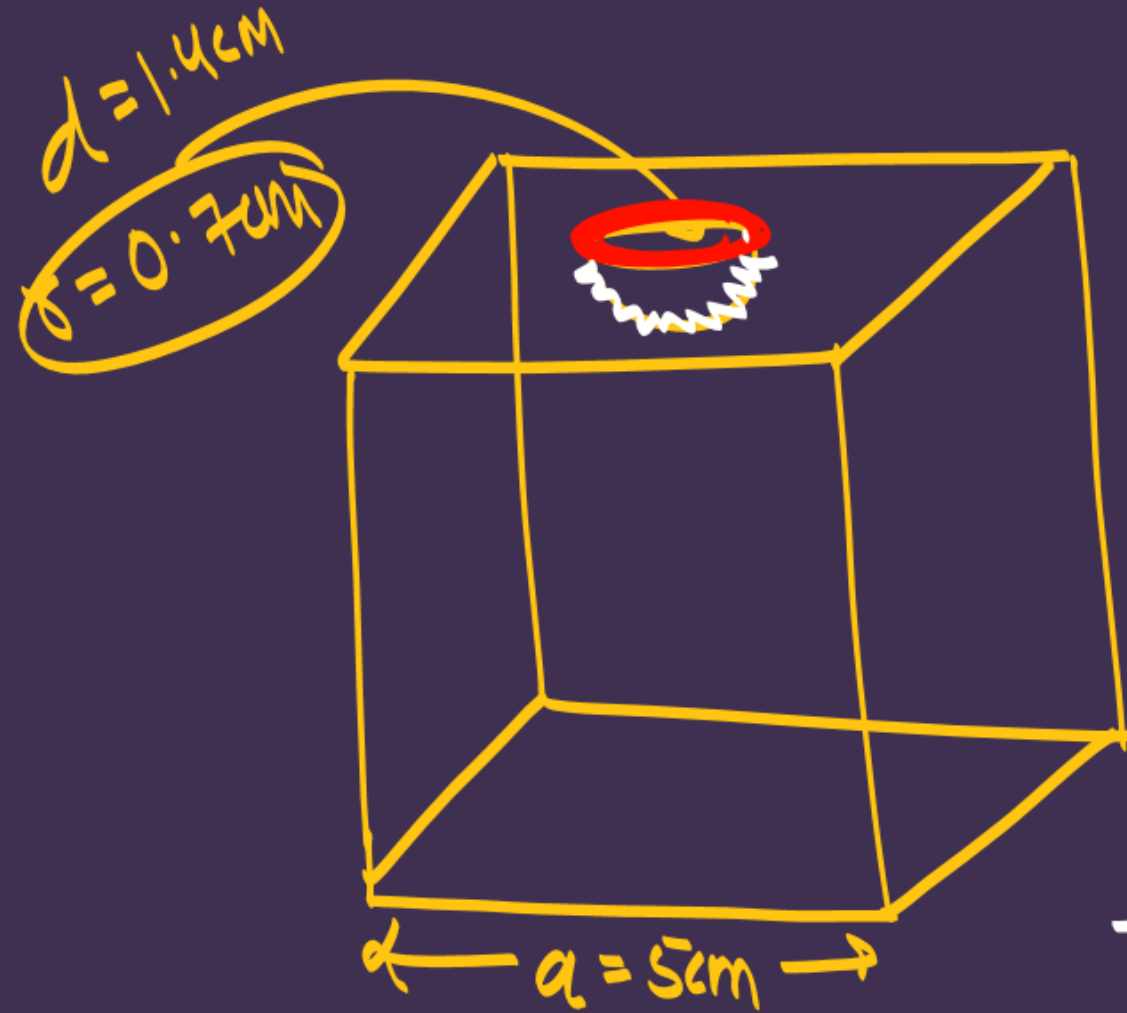
$$\Rightarrow \underline{\underline{\frac{2}{3}\pi r^3 \text{ mm}^3}}$$



$$SA \Rightarrow \left(\text{golf ball sphere SA} \right) - 315 \left(\text{ar. of circled dimple} \right) + 315 \times \left(\text{SA dimple hem.} \right)$$



#LP: A wooden cubical die is formed by forming hemispherical depressions on each face of the cube such that face 1 has one depression, face 2 has two depressions and so on. The sum of number of hemispherical depressions on opposite faces is always 7. If the edge of the cubical die measures 5 cm and each hemispherical depression is of diameter 1.4 cm, find the total surface area of the die so formed.



face 1 $\rightarrow$ 1 hem

2 $\rightarrow$ 2 "

3 $\rightarrow$ 3 "

4 $\rightarrow$ 4 "

5 $\rightarrow$ 5 "

6 $\rightarrow$ 6 "

Total hemispheres dug out $\Rightarrow$ 21 hem

[CBSE 2025]

TSA die $\Rightarrow$

$$\left[(\text{TSA cube}) - 21 \times (\text{ar of circle hem.}) + 21 (\text{CSA hem}) \right]$$

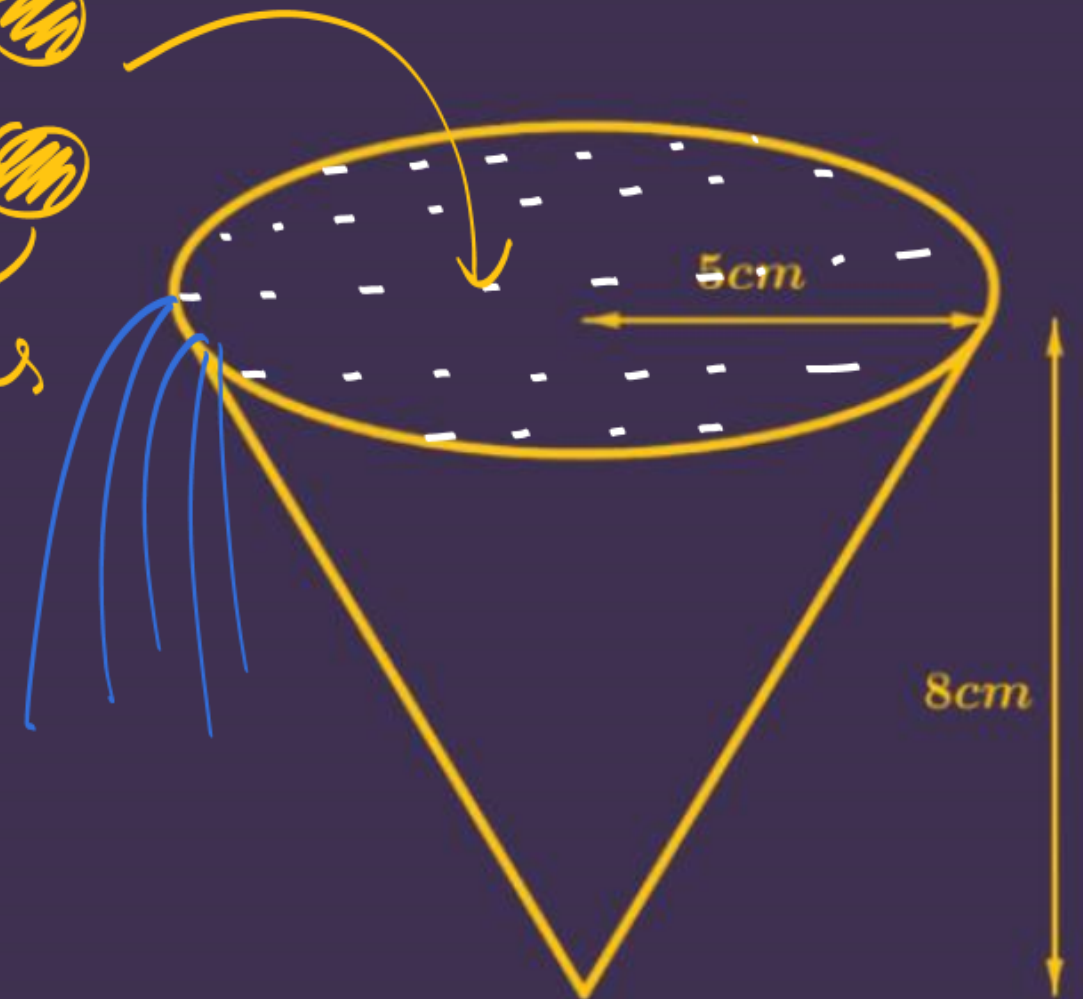
#LP: A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped in the vessel.

Vol. Water out = $\frac{1}{4} \times \text{Vol cone}$

$$x \times \left(\frac{4}{3} \pi (0.5)^3 \right) = \left[\frac{1}{4} \times \frac{1}{3} \pi (5)^2 (8) \right] \quad x \text{ 'spheres'}$$

$$x \times \frac{0.5}{10} \times \frac{0.5}{10} \times \frac{0.5}{10} = \frac{1}{2} \times 5 \times 5$$

$$\frac{x}{100} = 1 \rightarrow x = 100 \text{ spheres}$$

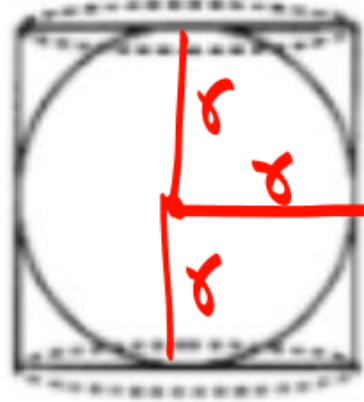


#LP: Directions: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).**
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).**
- (c) Assertion (A) is true but reason (R) is false.**
- (d) Assertion (A) is false but reason (R) is true.**

#LP:

Assertion (A): In the given figure, a sphere is inscribed in a cylinder. The surface area of the sphere is not equal to the curved surface area of the cylinder. ⓕ



Sphere $\rightarrow 4\pi r^2$

CSA of $\rightarrow 2\pi rh$
 $\rightarrow 2\pi r(2r)$
 $\times 4\pi r^2$

[CBSE SQP 2026]

Reason (R): Surface area of sphere is $4\pi r^2$ ⓕ

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- ✓ d) A is false but R is true.

#LP:

Singing bowls (hemispherical in shape) are commonly used in sound healing practices. Mallet (cylindrical in shape) is used to strike the bowl in a sequence to produce sound and vibration.



One such bowl is shown here whose dimensions are :

Hemispherical bowl has outer radius 6 cm and inner radius 5 cm.

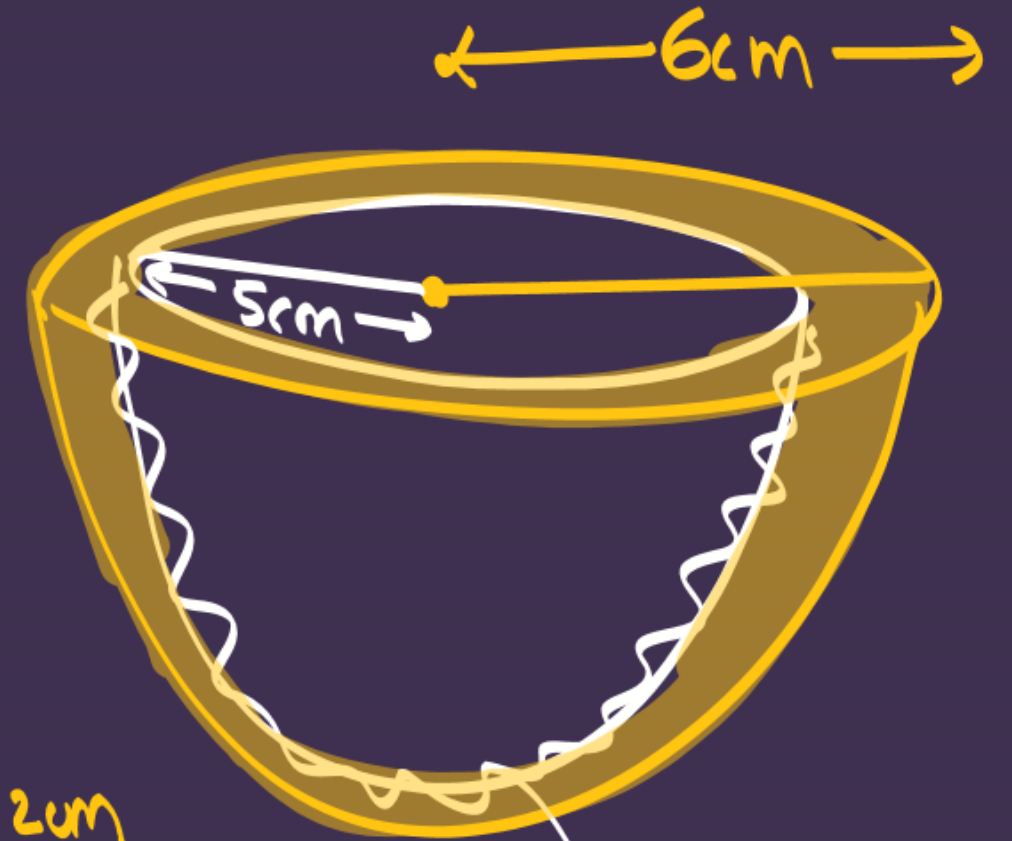
Mallet has height of 10 cm and radius 2 cm.

Based on the above, answer the following questions :

- (i) What is the volume of the material used in making the mallet ?
- (ii) The bowl is to be polished from inside. Find the inner surface area of the bowl.
- (iii) (a) Find the volume of metal used to make the bowl.

OR

- (iii) (b) Find total surface area of the mallet. (Use $\pi = 3.14$)



(i) Inner SA $\Rightarrow 2\pi r^2$
 $\Rightarrow 2\pi(5)^2 \text{ cm}^2$

(iii) Vol. of metal used in bowl
Vol. outer - Vol. inner
 $\Rightarrow \left[\frac{2}{3}\pi R^3 - \frac{2}{3}\pi r^3 \right] \text{ m}^3$

$V = \pi r^2 h$

6/14

Completed!

Thank you Goodies !!

